CHAPTER 7

THE HURRIAN LANGUAGE AS A LIKELY CANDIDATE:

PHONOLOGY

After we have observed, from a historical point of view, that presence of Hurrians in Crete would *a priori* not be inconceivable in the middle of and possibly even from the beginning of the second millennium B.C., it is time to see whether phonological and morphological features of the Hurrian language in general and of Hurrian onomastics in particular might be in accordance with features of the idiom written with Linear A. Some features can be deduced directly from Linear A, provided that the Linear A signs are provisionally transliterated with the phonetic values of their Linear B counterparts. Others can be deduced indirectly on the basis of orthographic incompatibilities of Linear B which cannot be explained from the phonology of the Greek language. It may be useful to start with the latter, as they are, in a sense, familiar to all students of Mycenaean Greek.

To be clear, when I use the term incompatibilities, I do not refer to the orthographic conventions of Linear B regarding the peculiar way in which, for instance, consonant clusters or diphthongs are represented, because these features merely arise from the fact that we are dealing with a script that only has at its disposal a range of signs representing open syllables. In this respect we should realize that this deficiency of the script is only a nuisance to *us,* when we try to understand information written with Linear B, but was probably no problem at all for the scribes who used the script, since they knew the repertoire of items they were describing and had no difficulty in reading what they had written. The orthographic incompatibilities of Linear B which concern us regard the representation of the occlusives and the liquids, as well as the unsatisfactory way in which the phoneme **[*h*]** is represented.

In Linear B we find no separate signs for the voiceless, voiced and aspirate stops, except for the dentals for which two series are distinguished, one voiced and one voiceless. These incompatibilities cannot be explained from Greek phonology, since distinction of voiceless ***π, τ, κ*** from voiced ***β, δ, γ*** and aspirate ***φ, θ, χ*** is phonemically significant in the Greek language. Since an alleged 'economy' of the writing system employed by the scribes to use no more signs than strictly necessary, does not provide a satisfactory explanation either, we may assume that this peculiar phenomenon had most likely been adopted - without alterations to accommodate the requirements of Greek phonology - from the system of Linear A which may have been designed for the notation of a language with a phonological system where opposition between voiceless, voiced and aspirate stops was non-phonemic. Theoretically Linear A, in its turn, might ultimately have adopted this phenomenon from the Cretan 'pictographic' script, but it seems wise first to examine whether Linear A itself might provide a coherent and satisfactory explanation without contaminating the discussion with speculation about an even smaller corpus of texts.

In the Hurrian language opposition between voiced and voiceless stops was non-phonemic. This is clearly shown by comparing Hurrian texts written by native speakers of Akkadian or West Semitic (Amorite and Ugaritic) with those written by native Hurrians themselves. Since voicing of occlusives was phonemic in Akkadian and West Semitic, native speakers of these languages were able to recognize the phenomenon of allophony in the Hurrian vernacular (cf. for this phenomenon, for instance, the allophonic differences between English and American pronunciation of /***t***/ in 'a little'). Thus, Hurrian texts from Ugarit (modern Ras Shamra) written in an alphabetic cuneiform have revealed a very consistent distribution of voiced and voiceless consonants. In them consonants become voiced when occurring singly after vowels and when adjacent to sonants. P.M. Purves, 'Hurrian consonantal pattern', *AJSL 58 (1941)*, 378-404, has discovered that this pattern also appears in Hurrian names in Akkadian documents written by Semitic scribes not only at Nuzi but at many other sites. He found this pattern also in the Hurrian texts from Mâri and in the Sumerian-Hurrian vocabulary from Ugarit and concluded that all the texts just mentioned, including those in alphabetic cuneiform, were written by Semitic scribes.

His conclusion was confirmed by the fact that the apparently Semitic scribe of Mâri No. 3 wrote 'pattern' Hurrian on the reverse and an Akkadian passage on the obverse in such a perfect way, both orthographicaly and grammatically, as none of the Hurrian scribes from Nuzi has succeeded in doing. The patterned treatment of Hurrian consonants by Semitic scribes, particularly when using the 'Babylonian' syllabary of Mâri or the alphabetic of Ugaritic, which faithfully distinguished the voiced and voiceless stops, surely reflected differences which they actually heard, but - and this is vital - *the differences were not significant to and were probably not even perceived by the Hurrians themselves* (cf. P.M. Purves, *NPN,* 184). Since voicing in Hurrian was non-phonemic and only allophonic caused by some contiguous voiced sounds, native Hurrians, when writing Akkadian, were not capable of recognizing consistently the phonemic differences between voiced and voiceless occlusives and used the signs for these stops indiscriminately, especially at sites such as Nuzi which was predominantly Hurrian, and Alalaḫ where Hurrian penetration was pronounced (cf. F.W. Bush, *A grammar of the Hurrian language,* Brandeis University 1964, 51-53).

We have observed that the orthographic incompatibilities in representing the occlusives in Mycenaean Greek, a phenomenon which Linear B probably inherited from Linear A, cannot be explained from Greek phonology, and we should add, nor from that of any other Indo-European language. However, from the facts just mentioned we should infer that they cannot be explained from Akkadian or West Semitic either. But Hurrian, where positional voicing only produced allophones, phonemically insignificant to its native speakers, provides the exact phonological explanation we have been looking for. One problem needs to be clarified. It concerns the representation of two separate series of syllabic signs for the voiced and voiceless dental stops in Linear B, one for ***τ-*** and ***θ-*** and one for ***δ-***. Judging from the repertoire of syllabic signs in Linear A, one may infer that Linear A has two separate series for the dental occlusives as well, and does in this respect correspond exactly with the Linear B repertoire, but not with the Classical Cypriot one, which has a single series for the dental stops as it has for the labials and dorsals, but which has two separate series for ***λ-*** and ***ρ-***, cf. M. Lejeune, *Mémoires de philologie mycénienne I*, Paris 1958, 327ff.; M. Pope - J. Raison, *Études Minoennes I*, BCILL 14 (1978), 29-31; D.W. Packard, *Minoan Linear A*, 116- 117.

We may conclude that Linear A scribes at different sites showed in general remarkable consistency with regard to the orthography of terms and names (the bulk of Linear A texts was found in the layers of destruction at the end of Late Minoan IB, dated to ca. 1450 B.C.), perhaps as much as their Linear B colleagues would display a few centuries later at Knossos, Mycenae, Pylos, Thebes ca. 1200/1180 B.C.. Most differences in orthography within Linear A and B can be explained. Some spellings may show idiosyncratic features due to the preferences of some scribes, but they pale into insignificance beside the different spellings displayed by scribes using the cuneiform syllabary.

In syllabic cuneiform it is in principle not easy, if not impossible, to establish, whether consonants are voiced or voiceless at the beginning of a sequence and in consonant clusters, because the clear indications of *double writing* for *voicelessness* and *single writing* for *voicing*, that are available in intervocalic position, are failing. In general one often assumes that Semitic scribes were more able to distinguish Hurrian voicing from voicelessness than Hurrian scribes, because voicing was positional and non-phonemic in Hurrian, with allophones as a result. Therefore the decipherment of the alphabetic cuneiform script of Ugarit came as a relief for many scholars, because that script showed a clear distinction between voiced and voiceless consonants and seemed to indicate that voicing at the beginning of Hurrian words and names did not exist at all.

According to Th. Richter (*BGH*, 461) ***teš-*** is the reconstructed root for forms such as *te-ša-ḫi*, *te-ši-ḫi*, *ti-iš-šu-uḫ-ḫi*, *te-šu-ḫa-ar-ḫi-ta*, *ti-i-šu-wa-ar-ḫa-a-am-ma*, *tešu-ḫarḫi*, *tešuḫi* as well as the theonym ***Teššup***. Alphabetic cuneiform ***tṯ*** corresponds with ***te-š*** in personal names. See also E. Laroche, *GLH*, 263, s.v. ***Teššub***: Alph. ***tšb*** ou ***tṯb***, RS *passim*; cf. *Ugar. V*, 539. E. Laroche, *GLH*, 263, s.v. ***tešuḫi*** 'chef': (1) RS *te-ša-ḫi*, *te-ši-ḫi* = sum. UGULA = akk. *aklu*; RS voc. III 9 1; cf. Thureau-Dangin, *Syria XII*, 260; Speiser, *IH*, 47, 50, 114; *CAD* A I, 277. (2) Alph. ***tžǧ***, RS 24.644, 10; 643 A 17; cf. *Ugar. V*, 517. (3) Bo. *te(š)šuḫi* : *te-(eš)-šu-ḫa-a-ri*, KUB XXIX 8 II 50, IV 32; pl. gén. *te-eš-šu-ḫi-na-ši*, KUB XXXII 26 II 9; *ti-iš-šu-uḫ-ḫi*, KUB XLVII 6 II 11. (4) *tešuḫarḫi*, KBo XX 131 II 22; 129+ III 4; IBoT II 39 Ro 45, Vo 5.

It is clear that the alphabetic cuneiform forms ***tšb*** and ***tṯb*** suggest that the theonym ***Te(š)šub*** contains a *voiceless initial* dental. However, in all onomastics with ***-te(š)šup*** as second element and with a first element ending in a vowel, the dental of ***-te(š)šup*** is written *with single writing* indicating that it is *voiced*, e.g. at Nuzi: ***Aki-tešup*** (wr. *A-ki-te-šup* (N.B. the form *A-ki-it-te-šup* may look like an exception, but it owes its ***-t-t-*** to the fact that it is the result of assimilation < *A-kip-te-šup* that is also attested), ***Akku-tešup*** (wr. *Ak-ku-te-šup*), ***Ari-tešup*** (wr. *A-ri-te-šup*), ***Aštari-tešup*** (wr. *Aš-ta-ri-te-šu-up*, *A-aš-ta-ri-te-šup*, *Aš-ta-ri-te-šup*), ***Eḫli-tešup*** (wr. *Eḫ-li-te-šup*, *Eḫ-li-te-eš-šup*), ***Ḫami-tešup*** (wr. *Ḫa-mi-te-šup*), ***Ḫišmi-tešup*** (wr. *Ḫi-iš-mi-te-šup*, *Ḫi-iš-me-te-šup*), ***Ḫui-tešup*** (wr. *Ḫu-i-te-šup*), ***Ḫutanni-tešup*** (wr. *Ḫu-ta-an-ni-te-šup*), ***Ikki-tešup*** (wr. *Ik-ki-te-šup*), ***Inzi-tešup*** (wr. *In-zi-te-šup*), ***Iriri-tešup*** (wr. *Ir-ri-ri-te-šup*), ***Itḫi-tešup*** (wr. *It-ḫi-te-šup*, *It-ḫi-te-eš-šu-up*), ***Kai-tešup*** (wr. *Qa-i-te-šup*, *Ka-i-te-šup*, *Ka-i-te-eš-šup*, *Ga-i-te-šup*,), ***Kerari-tešup*** (wr. *Ge-ra-ri-te-šup*), ***Kibi-tešup*** (wr. *Ki-bi-te-šup*, *Ki-bi-te-eš-šu-up*), ***Milki-tešup*** (wr. *Mil-ki-te-šup*, *Mil-ki-te-eš-šu-up*, *Mil-ki-*^{d}*te-šup*), ***Nai-tešup*** (wr. *Na-i-te-šup*), ***Niḫri-tešup*** (wr. *Ni-iḫ-ri-te-šup*, *Ni-iḫ-ri-ti-šup*, *Ni-iḫ-ri-te-eš-šu-up*, *Ni-ir-ḫi-te-eš-šu-up*), ***Nikri-tešup*** (wr. *Ni-ik-ri-te-šup*), ***Ninki-tešup*** (wr. *Ni-in-ki-te-šup*), ***Pai-tešup*** (wr. *Pa-i-te-šup*, *Ba-i-te-šup*, *Ba-i-ti-šup*), ***Šaḫlu-tešup*** (wr. *Ša-aḫ-lu-te-šup*, *Šá-aḫ-lu-te-*

šup), ***Šašu-tešup*** (wr. *Ša-šu-te-šup*), ***Šeḫali-tešup*** (wr. *Še-ḫa-li-te-iš-šup*), ***Šeḫli-tešup*** (wr. *Še-eḫ-li-te-šup*), ***Ši-laḫi-tešup*** (wr. *Ši-la-ḫi-te-šup*), ***Šilwa-tešup*** (wr. *Ši-il-wa-te-šup*), ***Šipki-tešup*** (wr. *Ši-ip-ki-te-šup*), ***Šukri-tešup*** (wr. *Šuk-ri-te-šup*, *Šu-uk-ri-te-šup*), ***Tai-tešup*** (wr. *Ta-i-te-šup*), ***Tarmi-tešup*** (wr. *Tar-mi-te-šup*, *Tar-mi-te-eš-šup*), ***Tuḫmi-tešup*** (wr. *Du-uḫ-mi-te-šup*), ***Tupki-tešup*** (wr. *Dup-ki-te-šup*), ***Umpi-tešup*** (wr. *Um-bi-te-šup*), ***Urḫi-tešup*** (wr. *Ur-ḫi-te-šup*), ***Zili-tešup*** (wr. *Zi-li-te-šup*), ***Ziriri-tešup*** (wr. *Zi-ri-ri-te-šup*), cf. P.M. Purves, *NPN*, 265-266.

The name of the Hurrian Weather God is ideographically ^{d}IM or ^{d}U, or (with phonetic complement ***-ub***) ^{d}IM^{up} or ^{d}U^{up}, cf. F. Gröndahl *Die Personennamen der Texte aus Ugarit* (*PNTU*), Rome 1967, 263. In alphabetic cuneiform personal names it is ***tt***, ***dṯb***, ***tṯb*** (alphabetic ***ṯ*** = syllabic ***š-***). It is useful to compare sentence names with the theophorous elements ***Teš(š)u(b)-***, ***-teš(š)ub*** at Ugarit with alphabetic cuneiform equivalents:

ttmd = cf. *Tešub-mati* [Ugarit], cf. *Teššu-madi*, *Tešub-mati* [Nuzi; *NPN*, 265],
ttyd = cf. *Teššaia*, *Teššeia*, *Teššuia* [Alalaḫ 149],
ttn , cf. *Teššun* [Alalaḫ 149],
agdṯb, cf. *Agi-tešub*,
agtṯb, cf. *Agit-tešub* (assimilation) < *Agip-tešub*,
akdṯb, cf. *Akku-tešub* [Nuzi, cf. Purves, *NPN*, 199, 265],
antṯb, cf. *Ana-tešub*,
artṯb, cf. *Ar-tešub*,
urġtṯb, cf. *Urḫi-tešub*,
kltṯb, cf. **Kil-tešub*, cf. element *Kil-*,
ṯdṯb, first element *ṯ-* unexplained according to F. Gröndahl, *PNTU*, Rome 1967, 264.
ṯrtṯb, cf. **Šar-tešub*, cf. first element *Šar-*,
tbtṯb, cf. **Tubb-tešub*, cf. element *Tubb-*, cf. F. Gröndahl, *PNTU*, Rome 1967, 264.

The evidence from Ugarit shows that names beginning with the element ***Teššu-***, ***Tešša-***, ***Tešše-*** contain a voiceless dental. If ***-tešub*** is the second element, the alphabetic form offers ***-dṯb*** with voiced dental (corresponding with ***/-dešub/***), if the dental is intervocalic as in ***akdṯb***, cf. ***Akku-tešub***, and ***agdṯb***, cf. ***Agi-tešub***. The form ***agtṯb*** can be explained as the result of assimilation: ***Agit-tešub*** < ***Agip-tešub***. However, the form ***urġtṯb*** instead of an expected ****urġdṯb*** (cf. ***Urḫi-tešub***) is not in accordance with the orthographic convention that single writing of an intervocalic consonant in the cuneiform syllabary indicates that it is voiced.

A common explanation for such an exception to the rule may well be that the scribe wrote the form ***urġtṯb*** on the analogy of the theonym itself and in accordance with the fact that the majority of names with the element ***-tešub*** contained ***-tṯb*** in alphabetic cuneiform. The phenomenon is known as *"restoration on the analogy of a corresponding form"*. It also explains why we often find forms written according to a phonetic law alongside with forms restored on the analogy of a cognate form. The form ***antṯb*** instead of expected ****andṯb***, if compared with ***Ana-tešub*** attested at Ugarit, also seems an exception to the rule, but it should be admitted that the alphabetic form may be compared with ****An-tešub*** as, for instance, ***artṯb*** is to be compared with ***Ar-tešub***, in which case the voiceless dental of ***-tṯb*** is probably due to the fact that it is part of a consonant cluster.

It is clear that the dental of ***-tṯb*** is voiceless in all other cases where it is part of a consonant cluster: e.g. ***kltṯb***, cf. ****Kil-tešub***; ***ṯrtṯb***, cf. ****Šar-tešub***; ***tbtṯb***, cf. ****Tubb-tešub***.

The name of ***^{d}Ḫé-bat***, the absolutive and vocative, KBo II 21, 14; KUB XXVII 38 II 9, etc. In alphabetic cuneiform ***Ḫbt***, CTA 166, 56, 60, 62; 170 a 6, 10; 171, 3,5; 176, 5. The ergative: *^{d}Ḫé-bat-uš*, KUB XXIX 8 III 47; *^{d}Ḫé-pa-du-uš*, KUB XXVII 42 Vo 11. The genitive: *^{d}Ḫé-pa-at-ti*, KBo XIX 129 Ro 33, etc. (< the unassimilated form * *^{d}Ḫé-pa-at-wi*, cf. [*^{d}Ḫé*]-*bat-wi$_{i}$* KUB XXVII 4. 5), cf. E.A. Speiser, *IH*, 63, § 82, and E. Laroche, *GLH*, 100-101. So alphabetic cuneiform ***Ḫbt*** showed that ***Ḫ-*** in ***^{d}Ḫé-bat*** was voiceless, whereas e.g. the theophorous onomastic element ***-ḫepa*** in the personal name ***^{f}Putu-ḫepa***, Hurrian name of the Hittite queen, wife of Ḫattušiliš III, is in all instances of the cuneiform syllabary written with single writing of ***-ḫ-*** designating a voiced velar spirant, which is in accordance with the orthography of the queen's name ***pdġb*** in the alphabetic cuneiform of Ugarit. One must admit that the contrast of ***pdġb*** with the orthography of ***ḫbt*** instead of ****ġbt*** of the theonym ***Ḫépét*** / ***Ḫebat*** in the alphabetic cuneiform of Ugarit, suggesting that ***Ḫ-*** is voiceless, is striking (cf. e.g. E. Laroche, *GLH*, 100-101, s.v. *Ḫebat*). But how should this observation be valued ? In many languages the process of restoration of an original form on the analogy of a cognate form is a very normal phenomenon, if a phonetic law causes anomalous differences between cognate forms. Consequently one can sometimes observe simultaneous coexistence of a form that is the result of the process of a phonetic law next to a form restored on the analogy of a cognate form.

One need not be surprised that the enthiusiasm of the scholars who had discovered the phonological evidence from the alphabetic cuneiform script of Ugarit, soon led to the assumption that this evidence could be applied to all places where Hurrian had been written and spoken, regardless of whether these texts were all written with syllabic cuneiform and were often much older than the alphabetic cuneiform texts from Ugarit. There is no trace in the academic discussions that the alphabetic cuneiform evidence might only reflect a *late* phonetic development in the *local* Hurrian dialect of Ugarit, providing no evidence at all that the same developments had occurred as well earlier and elsewhere. The problem is that all other places with a Hurrian population outside of Ugarit only offer evidence on account of syllabic cuneiform that does not provide conclusive evidence for the quality of initial stops and of those in consonant clusters with regard to voicing or voicelessness.

It is intriguing why an almost unanimous claim of alleged voicelessness of initial consonants in Hurrian in general seems to exist as if the evidence that could in fact only be confirmed for Hurrian at Ugarit in ca. 1200 B.C, was automatically applicable to the vast areas of Hurrian residence over a period of many centuries. To me there is no doubt that the Hurrian language must not only be valued in diachronic or historic perspective, but that there must have been dialectal divisions as well. E.A. Speiser has indicated dialectal differences between Boğazköy Hurrian and the idiom of the Tušratta letter that can no doubt be traced back to the still not discovered Chancellery at Waššukkanni.

As regards the occlusives, Linear A has, like Linear B, only separate series for the voiced and voiceless dentals and Linear A uses initial voiced dentals, which seems to contradict Hurrian alphabetic cuneiform evidence from Ugarit for initial consonants. Nevertheless abundant evidence points to Hurrian or (more likely) a Minoan-Hurrian dialect as the best candidate for identifying the idiom written with Linear A.

During my research the evidence of correspondences between Linear A and Hurrian was and still is constantly accumulating and all pieces of the big puzzle seem to fall in the right place. My conclusion is that the Hurrian dialect written with Minoan Linear A used voiced initial dentals and probably also voiced dentals in consonant clusters. Since Linear A had no separate series for voiced and voiceless labial and palatal occlusives, again in the same way as Linear B, conclusive evidence for initial voicing and voicing in consonant clusters fails for these stops in Linear A. Initial voiced dentals in Linear A may well offer a warning that it is not permissible to apply automatically evidence from Ugarit for an alleged lack of initial voicing in the Hurrian dialect of that city at ca. 1200 B.C. to Hurrian texts elsewhere, written exclusively with the cuneiform syllabary and often much older as well. More caution seems advisable.

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